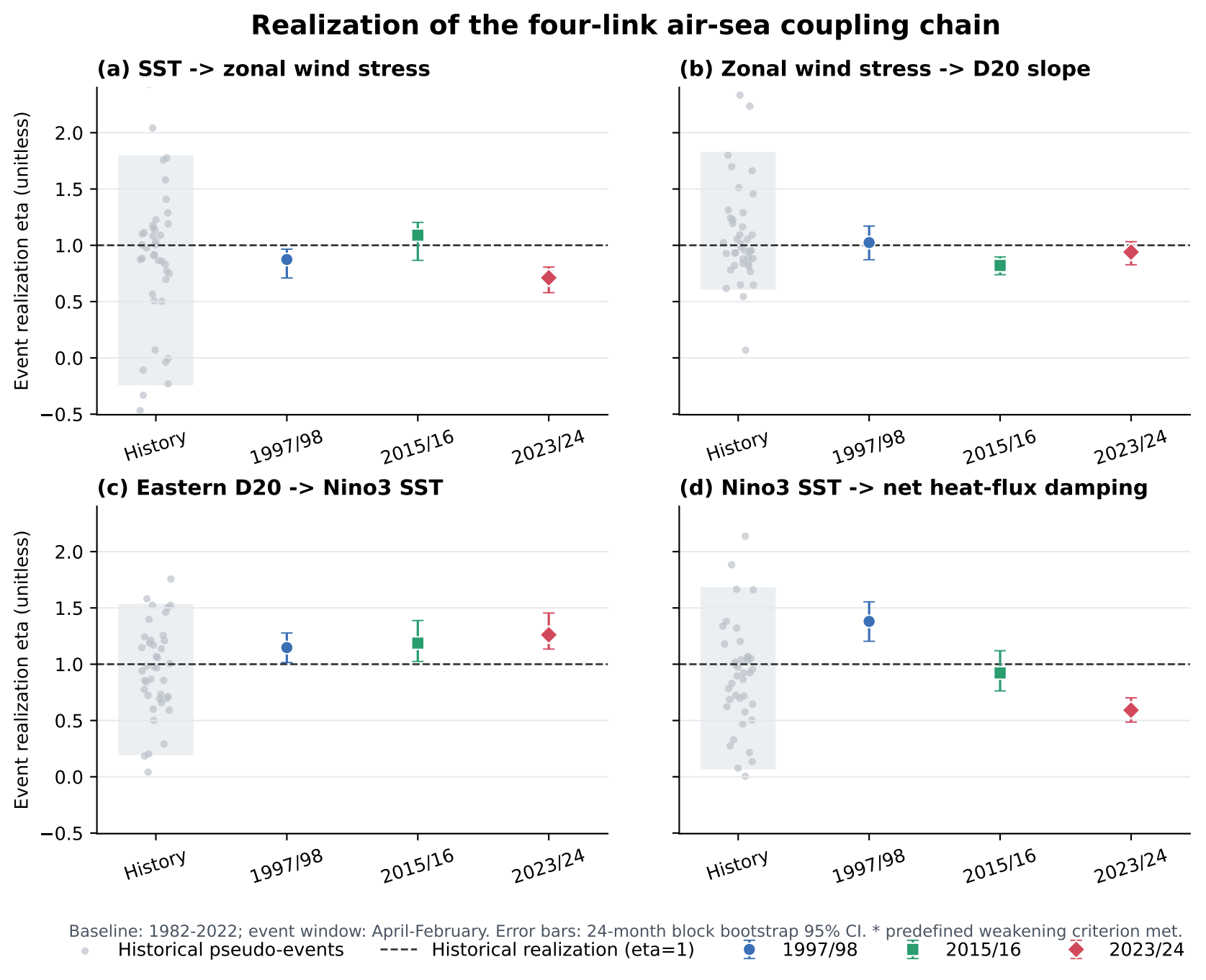
Realization of the four-link air-sea coupling chain
(a) SST -> zonal wind stress
(b) Zonal wind stress -> D20 slope
(c) Eastern D20 -> Nino3 SST
(d) Nino3 SST -> net heat-flux damping
Event realization eta (unitless)
2.0
1.5
1.0
0.5
0.0
−0.5
History
1997/98
2015/16
2023/24
Baseline: 1982-2022; event window: April-February. Error bars: 24-month block bootstrap 95% CI. * predefined weakening criterion met.
Historical pseudo-events
Historical realization (eta=1)
1997/98
2015/16
2023/24